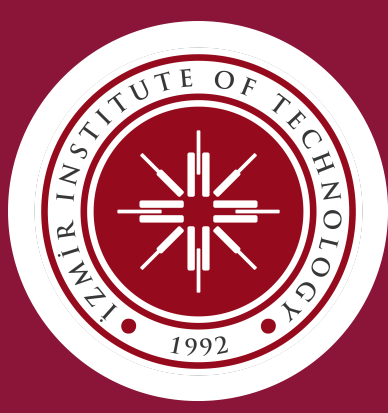




A Systematic Comparison of Large and Small Language Models in Standalone, Hybrid & Multi-Agent Architectures

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1 Abstract / Summary

Measurement-first platform benchmarking **ten architectures in three families** across **five public benchmarks** under a unified observability contract (accuracy · latency · cost · energy · carbon). We introduce **EATS** (Efficiency-Accuracy Trade-off Score) as an architecture-agnostic metric for comparing quality and efficiency together. Routing reaches **EATS 0.893** on HellaSwag and **0.854** on TruthfulQA; Pure Swarm leads GSM8K at **EATS 0.841**; Speculative achieves **EATS 0.908** on ARC. Novel **Blackboard-style** and **swarm** designs show task-specific strengths on reasoning benchmarks.

2 Background / Literature Review

LLMs offer stronger reasoning but impose higher latency, cost, and energy. **SLMs** are cheaper and easier to self-host, yet usually trail on broad reasoning tasks [1][2].

Our review of **54 collaborative SLM–LLM studies** shows a persistent gap: **accuracy is usually reported, but cost and energy rarely are** [10].

RQ1 Performance. How do collaborative systems compare with standalone LLMs on reasoning and classification?

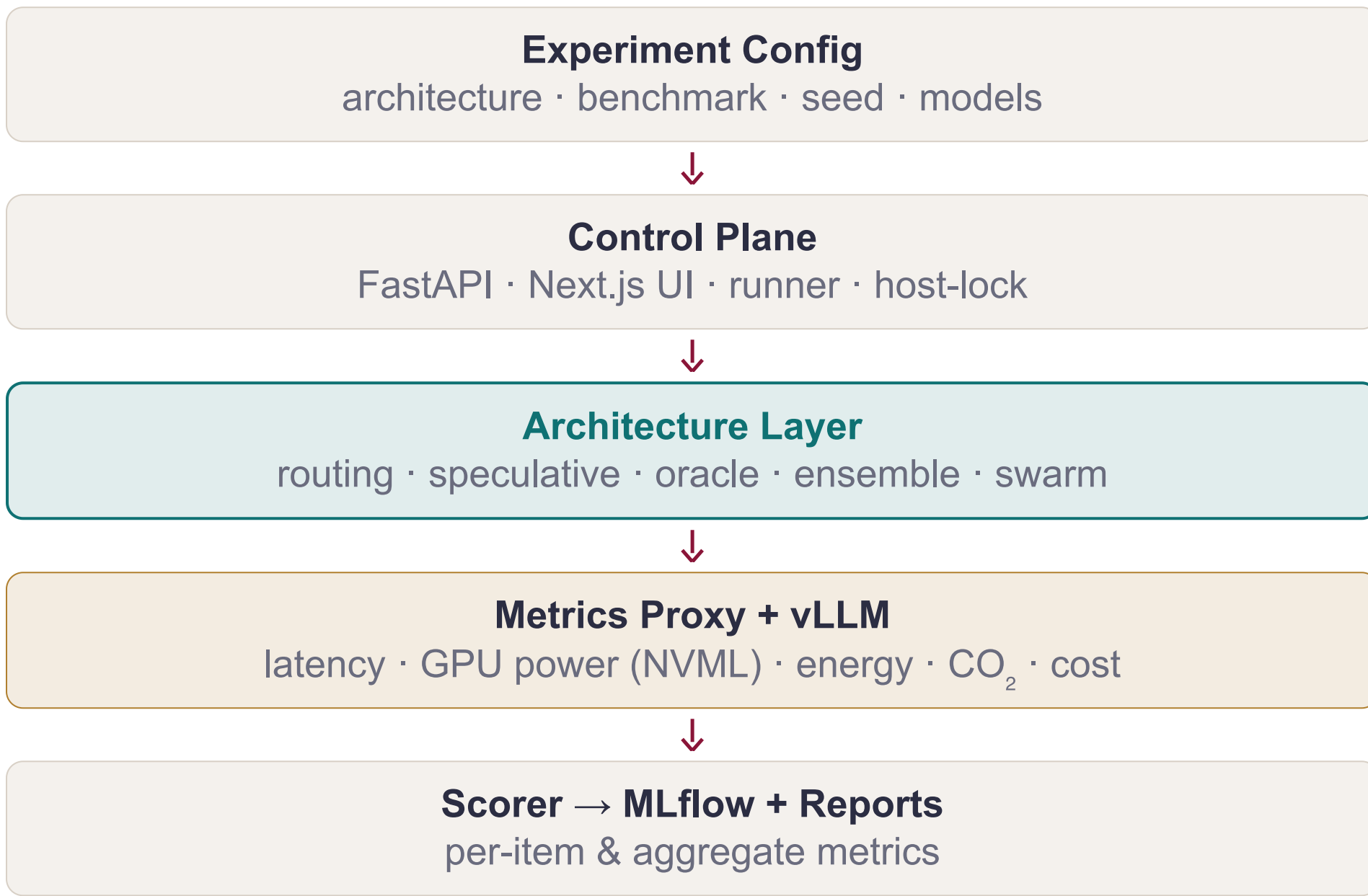
RQ2 Efficiency. How do hybrid and multi-agent designs change cost, latency, and energy?

RQ3 Unified metric. Does EATS provide a stable cross-architecture trade-off ranking?

RQ4 Empirical grounding. Do measured results match the trends from the 54-study SLR?

3 Measurement-First Platform

Every architecture consumes the same Query and returns the same Response carrying **accuracy, latency, tokens, cost, energy & CO₂** per query. The architecture layer is the *only* interchangeable component, so differences are attributable to orchestration logic rather than model API shape [4].



TIER	HOST	VRAM	MODELS
SLM	NVIDIA L4	24 GB	gemma4, qwen3.5, llama3.2 (3–4B)
Mid LLM	NVIDIA RTX PRO6000	96 GB	gpt-oss, gemma4, qwen3.5 (20–35B dense + MoE)
Heavy LLM	NVIDIA H200	141 GB	gpt-oss, llama3.3 (70–120B)

4 EATS · Efficiency–Accuracy Trade-off Score

A single score comparing any architecture against the standalone-LLM baseline.

$$\text{EATS} = \frac{\text{Accuracy}}{\text{Accuracy} + \lambda \cdot P_{\text{efficiency}} + \beta \cdot P_{\text{accuracy}}}$$
$$P_{\text{efficiency}} = 0.65 \cdot \text{Cost} + 0.20 \cdot \text{Latency} + 0.15 \cdot \text{Energy}$$
$$P_{\text{accuracy}} = 1 - \text{Accuracy}$$

$\beta = 0.60$ $\lambda = 0.40$

all terms normalised per standalone-LLM baseline · $\beta = 0.60$, $\lambda = 0.40$ calibrated via grid search on MMLU validation split

5 Energy, Carbon & Cost

Per-request measurements use GPU power sampling (NVIDIA NVML) at the inference host [4].

$$\text{Cost} = \text{GPU rate} \cdot \text{time}$$
$$E = P_{\text{GPU}} \cdot \Delta t$$

$$\text{Latency} = \Delta t \text{ per request}$$
$$\text{CO}_2 = k \cdot E$$

$k = 380 \text{ gCO}_2/\text{kWh}$ (EU avg) · E in joules + 3.6×10^6

6 Benchmarks & Protocol

Five benchmarks span the **capability dimensions most cited in our companion SLR**, letting us test whether efficiency gains persist as tasks shift from knowledge retrieval to chain-of-thought math and adversarial fact-checking [5][6][7][8][9].

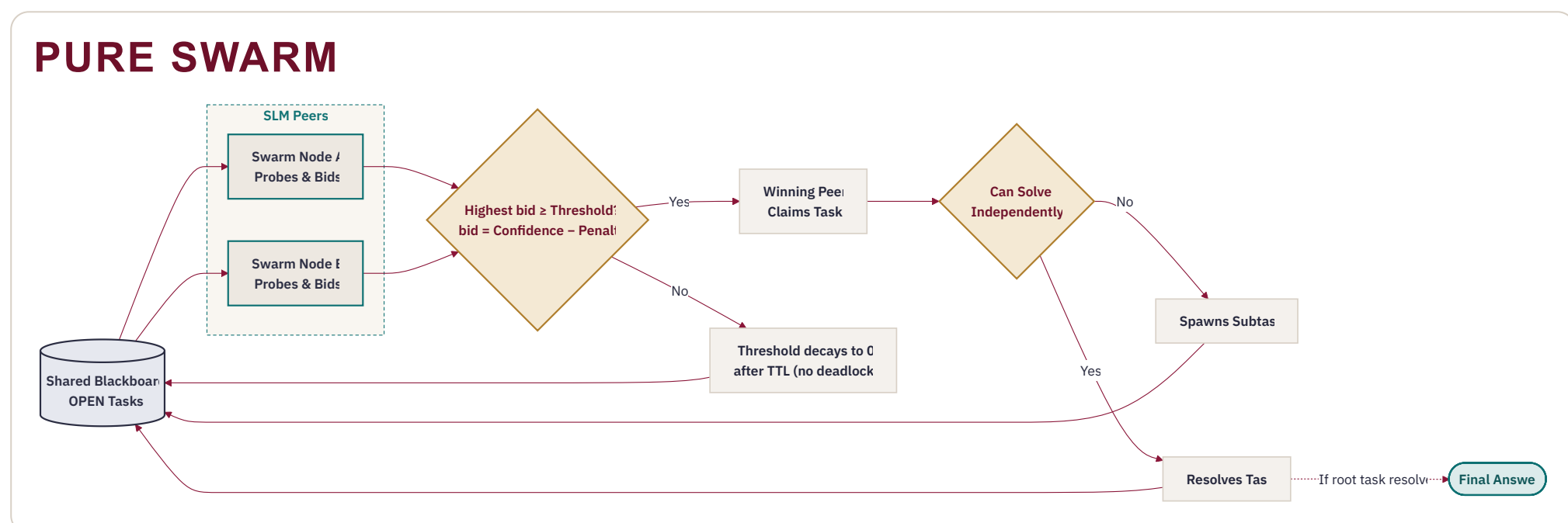
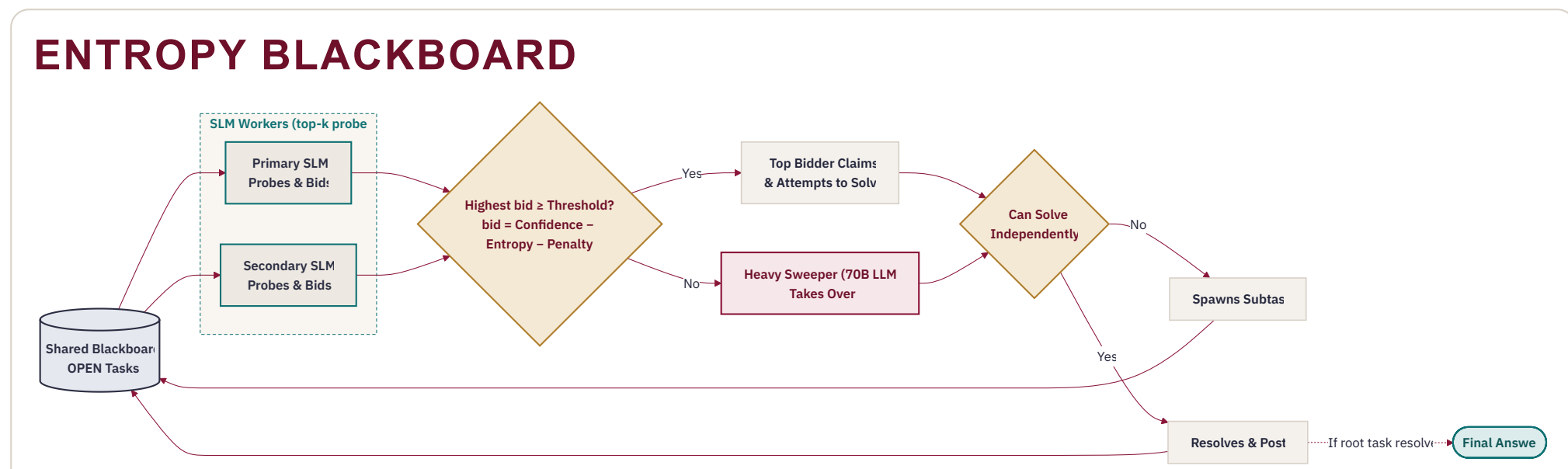
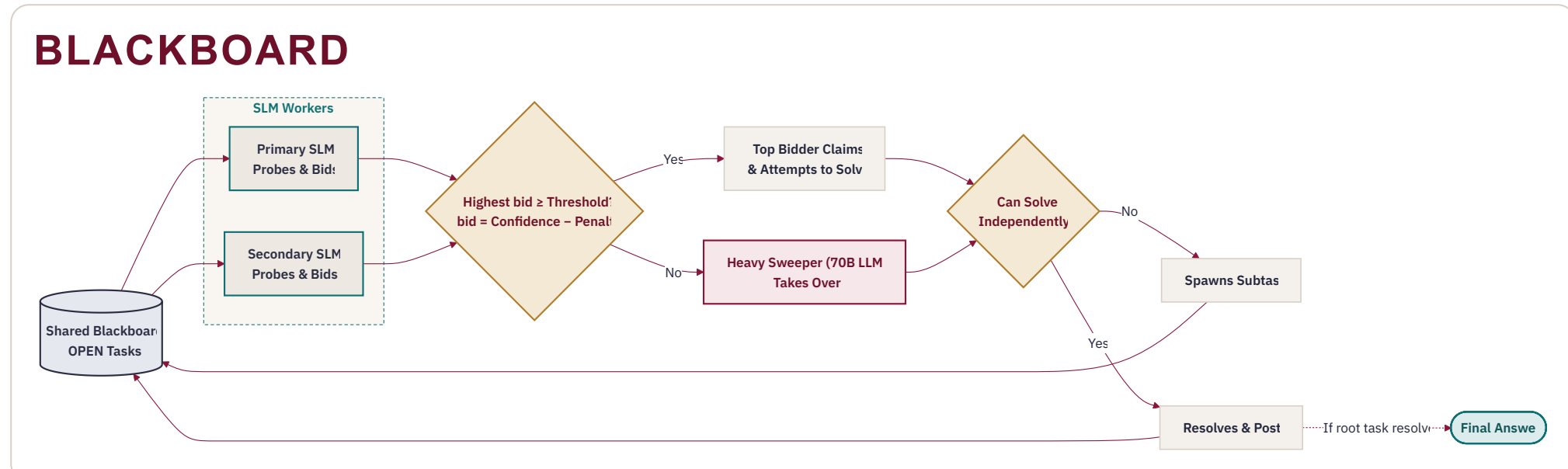
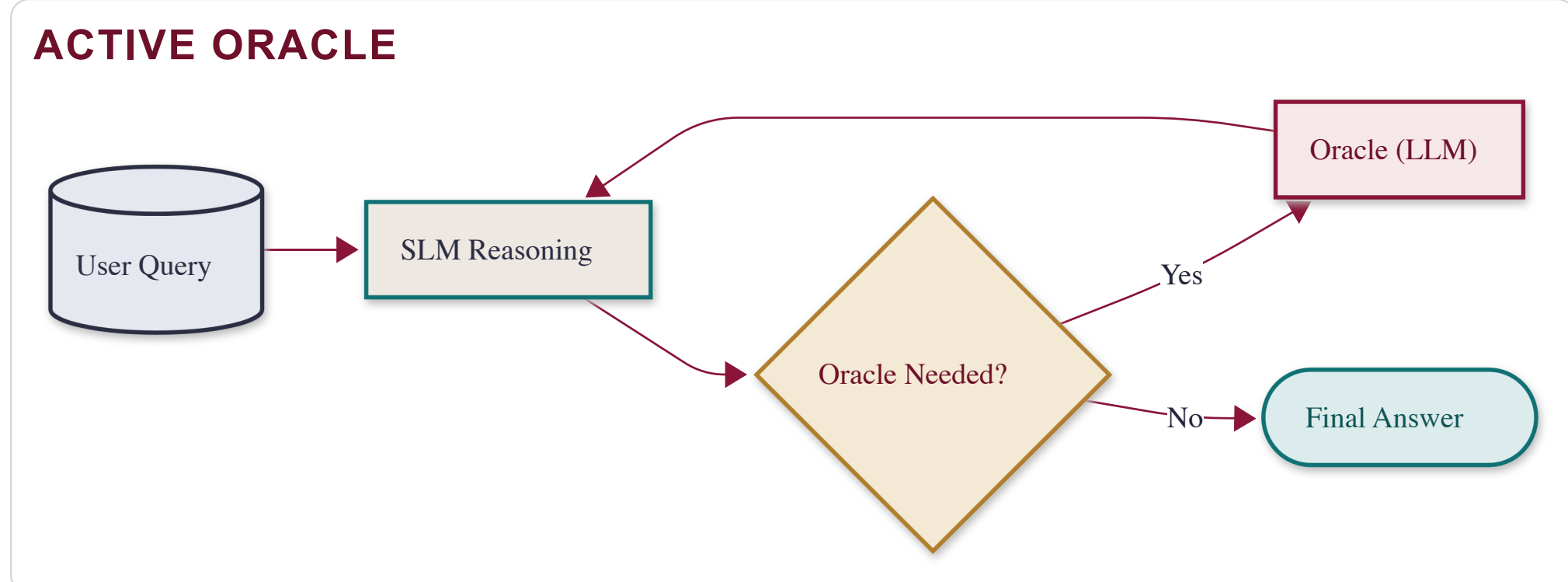
BENCHMARK	WHAT IT MEASURES
MMLU	Broad knowledge — 57 academic disciplines
ARC-C	Scientific reasoning — multi-step physical-world science
GSM8K	Mathematical reasoning — grade-school math, free-form
HellaSwag	Commonsense inference — sentence completion
TruthfulQA	Adversarial truthfulness — calibration vs. persuasive falsehoods

7 Architecture Taxonomy

Ten architectures in three families. † marks novel project contributions introduced in the current study.

The family progression runs from **internal sparsity** to **explicit coordination** [1][2][3].

FAMILY	ARCHITECTURE	KEY MECHANISM
STAND	Single-Model	direct inference
STAND	MoE	internal sparse expert activation
HYBRID	Routing	confidence SLM→LLM escalation
HYBRID	Speculative	SLM drafts; LLM rewrites suffix
HYBRID	Debate (POA)	proponent–opponent–arbitrator
HYBRID	Active Oracle †	targeted LLM sub-queries
HYBRID	Blackboard †	bid-based task board; LLM sweeper fallback
HYBRID	Entropy Blackboard †	uncertainty-penalised bidding
MULTI	Ensemble	majority or confidence-weighted voting
MULTI	Pure Swarm †	SLM-only peer coordination; no LLM online



8 Model Selection & Architecture Pairing

We keep the comparison focused on **architecture behavior** by mostly fixing the core hybrid pair: **Gemma4-4B** as worker and **Gemma4-31B** as verifier/sweeper.

SELECTION RULE Models were chosen for family diversity, open-weight reproducibility, and MoE coverage.	BEST-WORKING PAIRINGS Routing and Speculative with Gemma4-4B → Gemma4-31B gave the strongest EATS on ARC, HellaSwag, and TruthfulQA.
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P1 Routing / Speculative: Gemma4-4B → Gemma4-31B for the strongest quality–efficiency frontier.

P2 Oracle / Blackboard: Gemma4-4B workers + Gemma4-31B sweeper.

P3 Standalone MoE: Qwen3.5-35B-A3B and Gemma4-26B-A4B were the strongest efficient baselines.

9 UN SDGs

The thesis supports selected **UN Sustainable Development Goals** by making **cost, energy, carbon, and auditability** first-class outputs of architecture evaluation rather than afterthoughts.

SDG 8 **Economic efficiency:** cost-aware comparisons show when smaller or hybrid systems are sufficient, reducing unnecessary frontier-model spend.

SDG 9 **Open infrastructure:** the shared *Query/Response* contract and common reporting schema create reusable evaluation infrastructure.

SDG 10 **Reduced inequalities:** hybrid and multi-SLM results show that competitive AI quality can be approached without frontier-scale hardware or cloud budgets.

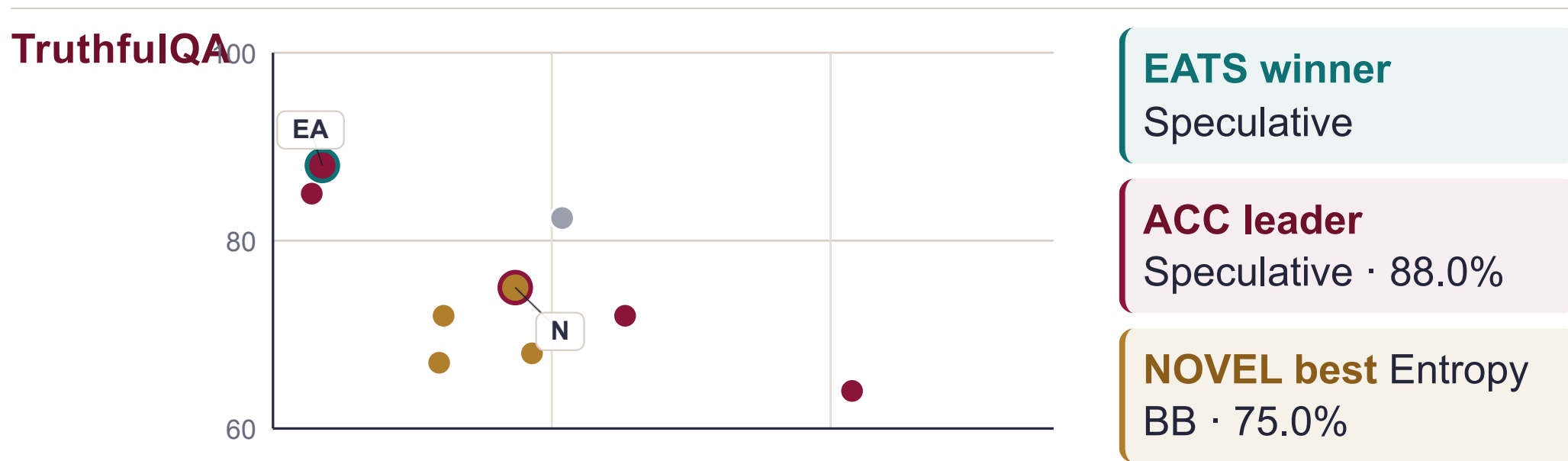
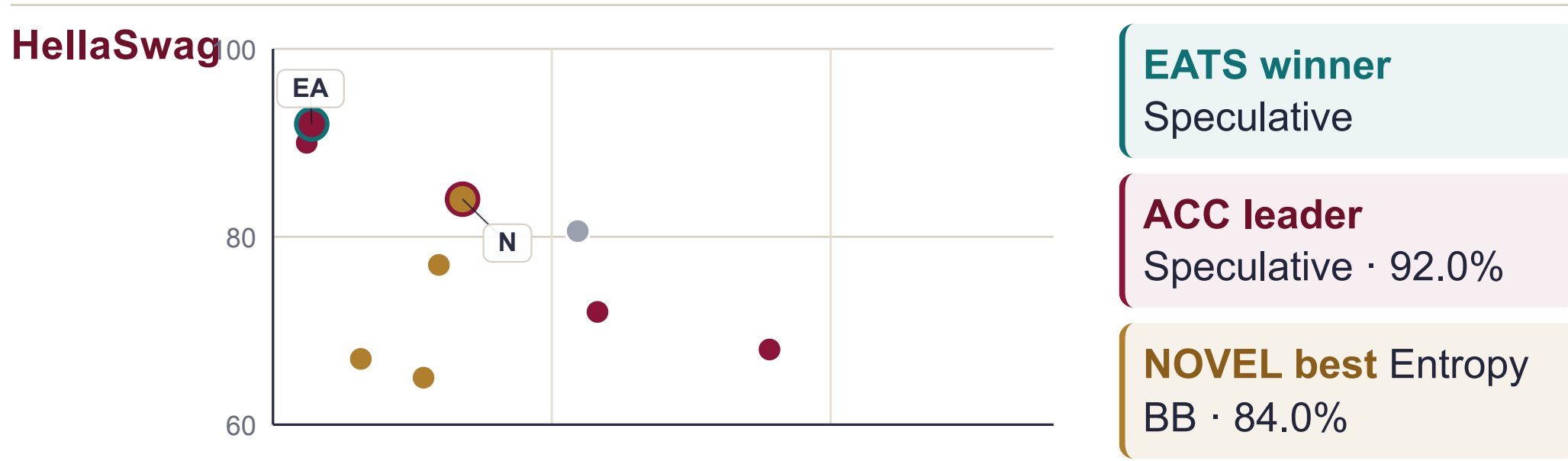
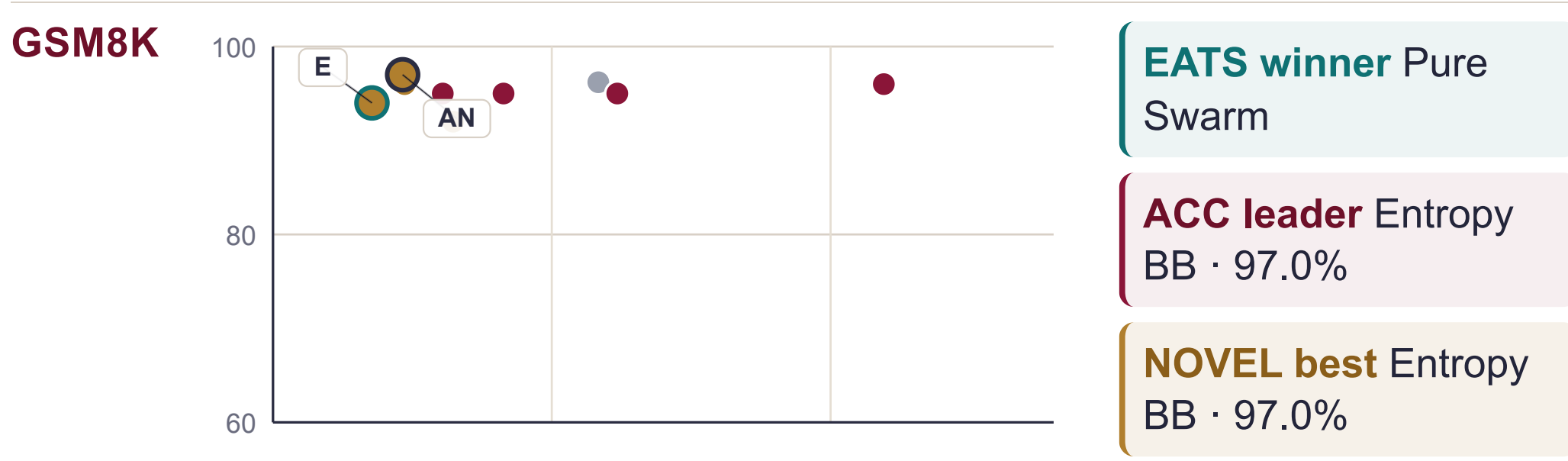
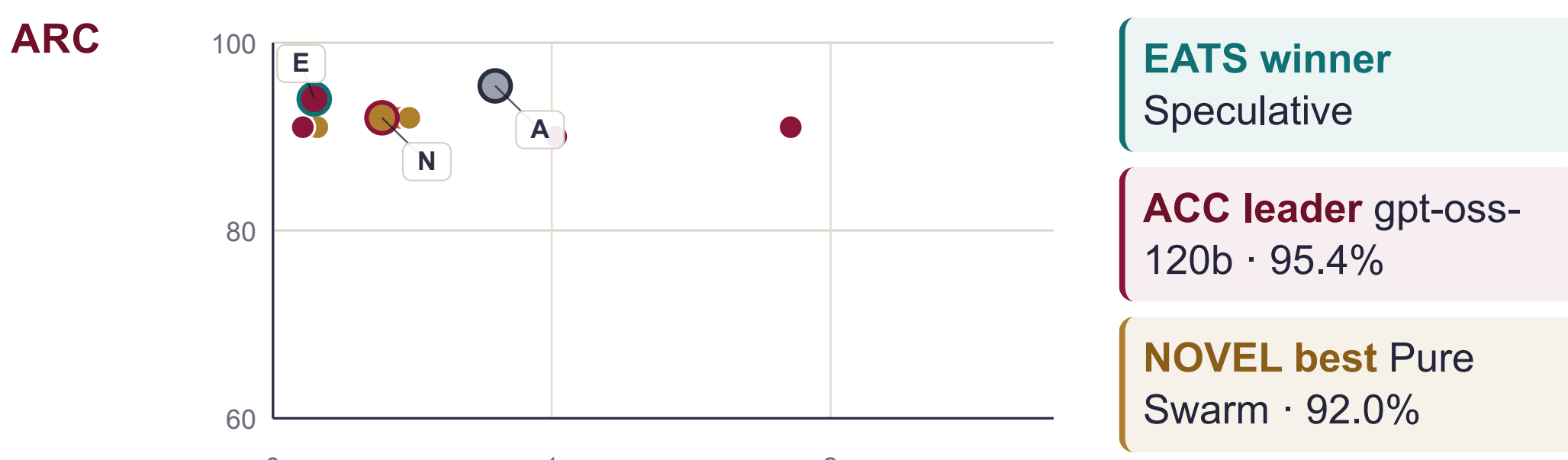
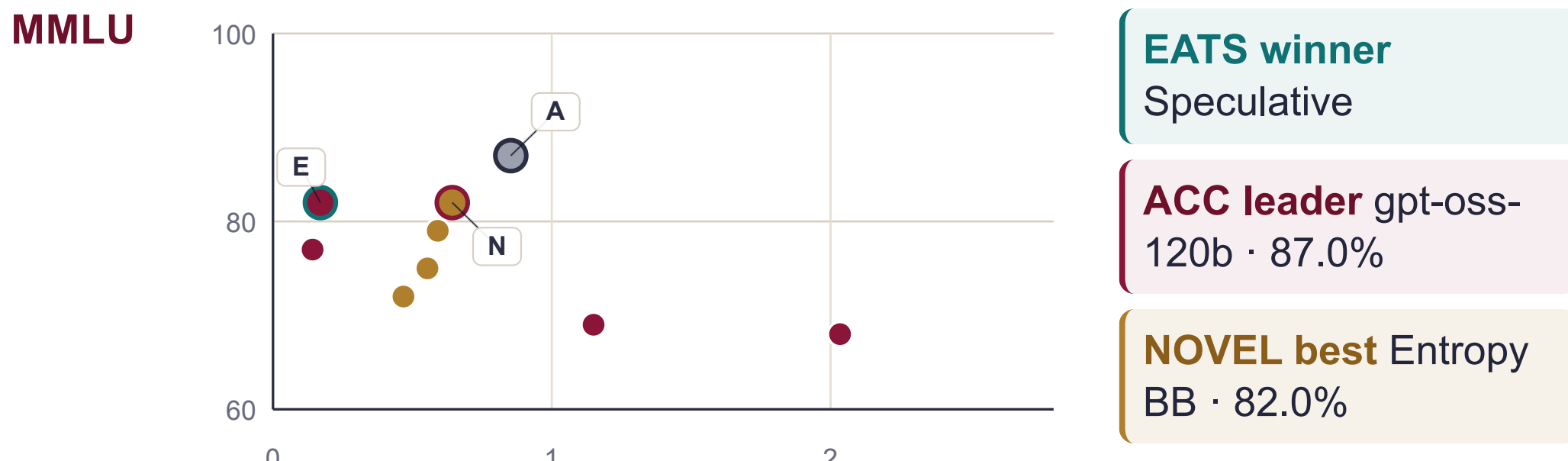
SDG 12/13 **Responsible energy use:** EATS includes energy and carbon in the main ranking, so wasteful paths are penalised directly.

SDG 16 **Accountable evaluation:** versioned benchmarks, host logs, and transparent metrics make architecture claims easier to audit.

The contribution is therefore **infrastructural**: it helps teams choose capable systems with clearer evidence about **quality, cost, and sustainability.**

10 Benchmark Readouts

Each mini-plot shows accuracy against efficiency penalty, so the strongest trade-offs appear toward the upper-left. **E** = EATS winner, **A** = accuracy leader, **N** = best novel result.



11 EATS Landscape

Canonical EATS for the latest qualified run of each architecture; the standalone row shows the best MoE baseline on each benchmark.

ARCHITECTURE	MMLU	ARC	GSM8K	HSWAG	TQA
Routing	0.80	0.90	0.78	0.89	0.85
Speculative	0.82	0.91	0.72	0.90	0.86
Standalone (MoE)	0.68	0.73	0.66	0.59	0.61
Blackboard †	0.69	0.80	0.82	0.67	0.55
Entropy BB †	0.69	0.79	0.83	0.69	0.60
Active O. †	0.67	0.89	0.75	0.67	0.64
Debate POA	0.52	0.66	0.64	0.53	0.52
Pure Swarm †	0.67	0.82	0.84	0.60	0.61
Ensemble	0.40	0.53	0.52	0.43	0.38

Darker = higher EATS · bold = benchmark winner · tinted rows / † = novel contribution.

12 Conclusions

The main synthesis is not that one model wins everywhere, but that **the best architecture depends on how much capability you can afford to escalate online.** That is why the poster reports both raw accuracy and EATS rather than collapsing the study into a single leaderboard.

RQ1 Collaborative systems are competitive with standalone LLMs; the strongest results come from routing, speculative decoding, and novel coordination on reasoning-heavy tasks.

RQ2 Hybrid escalation gives the clearest cost and energy gains, while multi-agent coordination usually pays a latency penalty.

RQ3 EATS gives a consistent cross-architecture ranking: speculative leads four benchmarks, and Pure Swarm leads GSM8K.

RQ4 The measurements corroborate 8 of 10 SLR trends: SLM-first escalation is the strongest general pattern, while coordination and entropy-aware bidding help on task-specific cases.

LIMITATIONS

- Single hardware stack (L4 / PRO6000 / H200) — results may not generalise to other GPU tiers or cloud inference endpoints.
- Benchmark scope covers 5 NLP task types; code generation, retrieval-augmented and long-context tasks are not included.
- EATS weights ($\beta = 0.60$, $\lambda = 0.40$) are fixed after calibration; sensitivity to alternative weight choices is not explored in this study.

In the full thesis, these findings will be extended with larger sample sizes, pair ablations, and broader task surfaces, but the poster already shows a stable message: **architecture design can buy back a large fraction of frontier-model quality without always paying frontier-model cost.**

SELECTED REFERENCES

- [1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Hendrycks et al. **MMLU**, ICLR 2021 · [6] Clark et al. **ARC Challenge Set**, 2018 · [7] Cobbe et al. **GSM8K**, 2021 · [8] Zellers et al. **HellaSwag**, ACL 2019 · [9] Lin et al. **TruthfulQA**, ACL 2022 · [10] Companion SLR (54 studies, PRISMA 2020).

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